

Siddarth Akalankam

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EDUCATION

Michigan State University

Bachelor of Science in Computer Science, Minor in Business

GPA: 3.76

East Lansing, MI

Aug. 2022 – May 2026

EXPERIENCE

Michigan State University × Amazon

Capstone Software Engineer

East Lansing, MI

Jan. 2026 – Apr. 2026

- Developed an observability platform that automated anomaly detection across multiple AWS services, accounts and regions, reducing manual monitoring for engineering teams by **2+ hours daily**.
- Built **Java** backend services and a serverless metric ingestion pipeline using **EventBridge**, **Lambda**, and **DynamoDB** to collect metrics across AWS accounts through cross-account **IAM role assumption**.
- Integrated **SageMaker** to execute time-series anomaly detection on live service metrics, automating the severity and priority ranking of incoming alerts.
- Designed a triage interface with direct **CloudWatch** links to specific timestamps, cutting investigation scope from **10,000+** logs to **50**.

Altair

AI Engineering Intern

Troy, MI

May 2025 – Aug. 2025

- Shipped a production API that enabled data scientists to deploy ML models seamlessly to both Azure cloud infrastructure and NVIDIA Jetson edge devices.
- Reduced inference latency from **4s to 100ms** by migrating serving from CPU to GPU with **TensorRT**, implementing in-memory model caching, and prewarming inference workers to eliminate cold-starts.
- Integrated the platform into Altair's existing microservice architecture using **RabbitMQ** for async communication, **Docker** for containerization, and **Kubernetes** for orchestration.

YouLearn AI

Software Engineering Intern

Remote

Sep. 2023 – Dec. 2023

- Reduced API latency by **40%** in a study material chatbot built with **OpenAI API** by implementing **Redis** caching for frequent vector queries.
- Designed **MongoDB** schemas for conversation history and containerized backend services with **Docker**, deploying to **Google Cloud Run** to autoscale conversational workloads.
- Developed automated integration and E2E test suites using **Pytest** to validate context retrieval accuracy, database queries, and LLM orchestration workflows.

PROJECTS

FlashML | *Next.js, FastAPI, ONNX Runtime, AWS ECS, AWS S3, PostgreSQL, RunPod Serverless*

- Built a SaaS platform that enables developers to deploy machine learning models as **production-ready APIs** through a single upload, **without managing compute or storage infrastructure**.
- Implemented a versioned model registry, API key authentication, and an automated inference pipeline that validates access and routes requests to CPU or GPU runtimes.

FinForge AI | *React, TypeScript, FastAPI, Plaid, Claude API, AWS Lambda, AWS SQS, PostgreSQL, Pinecone*

- Built a **personal finance AI agent that lives in the user's inbox**, syncing live banking data through Plaid, generating personalized financial briefings, and delivering them through **scheduled email workflows**.
- Engineered an agentic retrieval system that dynamically retrieves **live financial data, long-term memory**, and external tools to answer follow-up financial questions through email conversations.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL

Frameworks: React, Next.js, Express, FastAPI, Spring Boot

Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes, Terraform, CDK, RabbitMQ, Git, CI/CD

Databases: PostgreSQL, DynamoDB, MongoDB, Pinecone, Redis

AI/ML: PyTorch, ONNX Runtime, TensorRT, SageMaker, RAG